Survey on "Software Artifacts and Static Analysis Tools for Security Assessment" (PART 1)

Thank you for your interest!

This online survey is part of an empirical study conducted by the [redacted]. The goal of this study is to gain insights into the use of architectural rules (i.e., best practices) to enhance the security of systems based on Artificial Intelligence (AI) components. Particularly, into practitioners' views regarding the adoption and conformance checking of those rules. The study is divided into two parts (rewarded separately) and should take, altogether, around 13 min.

PART 1: Technical Assessment (**0.3 GBP reward**)

- It consists of 3 generic questions about programming, 1 related to software security and 1 to ML/AI.
- It takes around 3 min.
- Your answers will determine **if you meet our criteria** to undergo the 2nd part of the study.
- If you meet our participation criteria, you will be **invited to take the second part** of the survey.

PART 2: Main Survey (**1.3 GBP reward**)

- A set of questions about your experience securing AI-based systems.
- It takes around 10 min.

IMPORTANT:

1. To participate in this study, you must be at least 18 years old.
2. Please read the **Participant Information Sheet** before starting with the survey. The sheet can be accessed via this link: [redacted]

If you would like to proceed with the **Technical Assessment (PART 1)**, please answer the following questions by placing a check mark accordingly.

Question attribute	Value
hide_tip	1
other_position	end
save_as_default	N

I am age 18 or older

I have read and understood the Participant Information Sheet

I agree to participate in this research and I want to continue with the survey

Consider the following pseudocode that merges two unordered lists of characters in a specific way and prints the result:

```
list1 = ['X', 'K', 'M', 'Y']
list2 = ['Z', 'A', 'N', 'B']
merged_list = []
for i from 0 to length(list1) - 1 do:
    if list1[i] < list2[i]:
        merged_list.append(list1[i])
        merged_list.append(list2[i])
    else:
        merged_list.append(list2[i])
        merged_list.append(list1[i])
print(merged_list)
```

Question attribute	Value
save_as_default	N

What will be the output of this code?

(Tip: Choose one of the following answers)

Question attribute	Value
answer_order	normal
other_position	default
save_as_default	N

['X', 'Z', 'A', 'K', 'M', 'N', 'B', 'Y']

['Z', 'X', 'A', 'K', 'M', 'N', 'B', 'Y']

['Z', 'X', 'K', 'A', 'N', 'M', 'Y', 'B']

['X', 'Z', 'K', 'A', 'N', 'M', 'Y', 'B']

None of the above

Consider the following pseudocode that processes two unordered lists of numbers using a stack.

```
list1 = [18, 7, 11, 22]
list2 = [5, 13, 9, 16]
stack = []
for i from 0 to length(list1) - 1 do:
    if (list1[i] % 2 == 0):
        stack.push(list1[i])
    else:
        stack.push(list2[i])
result = []
while stack is not empty do:
    result.append(stack.pop())
print(result)
```

Question attribute	Value
save_as_default	N

What will be the output of this code?

(Tip: Choose one of the following answers)

Question attribute	Value
answer_order	normal
other_position	default
save_as_default	N

[18, 13, 9, 22]

[16, 9, 13, 5]

[22, 9, 13, 18]

[5, 13, 9, 18]

None of the above

Please look at the code snippet below to answer the questions that follow

```
main {
    print(func("hello world"))
}

String func(String in) {
    int x = len(in);
    String out = "";
    for (int i = x - 1; i >= 0; i--) {
        out.append(in[i]);
    }
    return out;
{
```

Question attribute	Value
save_as_default	N

What is the parameter of the function?

(Tip: Choose one of the following answers)

Question attribute	Value
answer_order	normal
other_position	default
save_as_default	N

String out

String in

I don't know

int i=x-1

i>=0

i--

Outputing a String

int x=len(i)

Please select the returned value of the pseudocode above

(Tip: Choose one of the following answers)

Question attribute	Value
answer_order	normal
other_position	default
save_as_default	N

hello world

hello world 10

dlrow olleh

world hello

HELLO WORLD

I dont't know

hello world hello world hello world hello world

Which of the following features have you previously added to your code?

(Tip: Select all that apply)

Question attribute	Value
other_replace_text	Other (please specify):
other_position	end
save_as_default	N

API keys

Authorization / Authentication

Encryption / Decryption

Fraud prevention

Input validation

Key management and generation

Secure network connections and communications

Signature generation and validation

Store and authenticate user names and passwords

Transfer files securely

I did not use any of these features

In a typical ML/AL pipeline, in which part of the dataset do you apply data balancing techniques (e.g., oversampling, undersampling)?

(Tip: Choose one of the following answers)

Question attribute	Value
answer_order	normal
other_position	default
save_as_default	N

Validation set

Test set

Training set

Adjustment set

Entire dataset

Thank you for your interest in our survey. Unfortunately, you do not meet some of the requirements to participate in this study:

- Be at least 18 years old.
- Read the consent form or had it read to you.

- Be comfortable using the English language.
- Agree to participate in this research.

If you have any questions or inquiries, please do not hesitate to contact us a [redacted]

Thank you for your response!

IMPORTANT: Click **"SUBMIT"** to record your answer!

Question attribute	Value
save_as_default	N